

Apple Support Twitter AI Agent

Technical Report

1. Problem Framing

Apple Support handles technical complaints at scale on a public platform. A good agent must do three things: **classify intent accurately**, **draft a reply that sounds like Apple** — empathetic, professional, Twitter-length — and **decide whether to escalate to a human agent** with a stated reason.

The dataset is **80,491 reconstructed conversation threads** from the AppleSupport Twitter account, collected during October–November 2017, immediately after the iOS 11 launch. Over 35% of messages concern iOS update issues or the autocorrect bug where the letter "l" autocorrects to "A?". Any system trained and evaluated on this data inherits that context.

What we chose not to build

- **No multi-turn conversational agent.** Apple Support's public Twitter workflow is single-turn — one customer tweet, one Apple reply. The DM conversation is out of scope.
- **No fine-tuning.** With 242 labelled examples and 10 intents, fine-tuning would underfit. Few-shot prompting is the right tool for this problem size.
- **No production serving infrastructure.** This is a research prototype demonstrating the pipeline is sound.

2. Intent Taxonomy — Discovered, Not Assumed

Rather than defining intents by assumption, we embedded all **76,656 customer messages** using **all-mpnet-base-v2** (768 dimensions), reduced with UMAP (→ 30 dims), and clustered with HDBSCAN (min_cluster_size=100). 106 raw clusters were merged into **10 final intents** based on one rule: two clusters belong to the same intent if Apple's troubleshooting reply would be identical.

The clustering revealed **autocorrect_i_bug** (13,414 messages, 17.5% of traffic) — a category completely invisible to keywords. Customers described this iOS 11 bug using "question mark", "boxes", "letter l", and the Unicode variation selector l. No keyword would have caught it. It is now the second-largest intent in the taxonomy.

Intent	Messages	% of corpus
general_inquiry	20,723	27.0%
ios_update_general	13,891	18.1%
autocorrect_i_bug	13,414	17.5%
app_issue	11,822	15.4%
device_hardware	4,988	6.5%
battery_drain	4,077	5.3%
account_access	3,179	4.1%
phone_freezing	1,807	2.4%
wifi_bluetooth	1,770	2.3%
order_purchase	1,170	1.5%

3. System Architecture

Four components run per message. **Retrieval/drafting and escalation run in parallel** after classification completes, saving ~1–2 seconds per message.

- **Classifier** — chain-of-thought prompting with 5 examples per intent. LLM reasons first ("primary problem: ..."), then outputs the intent name. Explicit boundary rules resolve the hardest cases.
- **Retriever** — filters 80,483 indexed threads to the classified intent, ranks by cosine similarity using per-intent embedding matrices. Returns 3 most semantically similar historical threads.

- **Reply drafter** — given message, intent, and 3 retrieved threads, drafts a 2–3 sentence Twitter reply grounded in real Apple Support responses.
- **Escalator** — two-layer decision: hard regex word-boundary triggers for unambiguous cases (legal threats, safety, crisis), then LLM judgment for ambiguous cases. Returns AUTO_HANDLE or ESCALATE with reason.

4. Results vs. Baselines

Evaluation set: **242 hand-labelled examples**, stratified across 10 intents (~24 per intent), sampled from the clustered corpus and independently verified. Human override rate: **26.4%** — the labeller disagreed with the clustering on 64 of 242 examples, confirming independent ground truth.

System	Accuracy
Random (always predict app_issue)	~14%
Keyword matching (trivial baseline)	57.9%
ComplementNB — 76k cluster labels	65.7%
Logistic Regression — 76k cluster labels	69.8%
LLM agent (this system)	82.2%

The LLM beats the best ML baseline by **+12.4 pp** and the keyword baseline by **+24.3 pp**.

Per-intent accuracy:

Intent	Correct	Accuracy
phone_freezing	17/17	100.0%
autocorrect_i_bug	24/25	96.0%
wifi_bluetooth	20/21	95.2%
app_issue	30/33	90.9%
device_hardware	14/16	87.5%
battery_drain	22/26	84.6%
general_inquiry	15/19	78.9%
order_purchase	23/30	76.7%
account_access	16/23	69.6%
ios_update_general	18/32	56.2%

Reply quality (LLM-as-judge, n=135): Relevance 4.31 · Tone 4.90 · Actionability 4.35 · Conciseness 5.00 · **Overall 4.56 / 5**

Human-judge agreement (30 replies): Overall correlation 0.687 · Within 1 point on 30/30 (100%) for tone, conciseness, overall · **Relevance diverges most** (judge 3.10 vs human 4.20 avg — judge penalises generic DM replies more harshly).

Escalation rate: **9.5%** (23 / 242 messages escalated to human agent)

5. Failure Analysis

43 of 242 predictions were wrong. Five most informative patterns:

F1 — ios_update_general is the weakest intent (56.2%, 14 wrong)

This is the catch-all for update complaints without a specific symptom. When a message names a specific consequence — Safari broken, WiFi failing, phone freezing — the LLM correctly routes to the specific intent, which counts as wrong against the label.

"Safari won't even open after installing the latest update on Mac OS" → True: ios_update_general / Predicted: app_issue

"whenever I try to update it says I'm not connected to the internet when I am" → True: ios_update_general / Predicted: wifi_bluetooth

Hypothesis: the "update caused X" vs "X" boundary was inconsistently labelled. A sharper rule — use the specific intent when a symptom is named, ios_update_general only for general dissatisfaction — would resolve most of these 14 failures.

F2 — Domain context the LLM cannot infer (account_access, 30% error rate)

All 6 account_access → general_inquiry failures involve phishing/scam emails with minimal context.

"hello. Is this legit?" → True: account_access / Predicted: general_inquiry

"where do I send spam Apple store emails?" → True: account_access / Predicted: general_inquiry

Hypothesis: a human Apple Support agent immediately recognises these as phishing questions from domain experience. The LLM sees three vague words and classifies on surface content. Adding phishing-specific few-shot examples would give the LLM the domain prior it is missing.

F3 — Apple Music contamination in order_purchase (76.7%, 7 wrong)

All 5 order_purchase → app_issue failures involve Apple Music subscription complaints — a direct consequence of merging cluster 20 (Apple Music) into app_issue during taxonomy construction.

"Getting tired of tracks randomly disappearing from purchases" → True: order_purchase / Predicted: app_issue

Hypothesis: "subscription", "purchased", "9.99" signal order_purchase; "Apple Music", "tracks", "songs" signal app_issue. A dedicated apple_music intent would resolve all 5 of these failures.

F4 — Multi-complaint messages split the LLM's attention

"the battery is shit and it keeps freezing since I updated" → True: battery_drain / Predicted: phone_freezing

Hypothesis: when a customer reports two symptoms, the LLM routes to whichever is most semantically vivid. "Keeps freezing" outweighs "battery is shit". This is inherent to single-label classification — the right answer depends on which problem Apple Support would prioritise, which is not always obvious from the message.

F5 — Escalation rate is unvalidated

The 9.5% escalation rate has no ground truth. Frustrated customers using hyperbole ("I'm going to lose my mind", "I swear I'm done with Apple") can pattern-match escalation triggers without containing genuine threats. The LLM second layer is meant to catch these, but without labelled ESCALATE/AUTO_HANDLE examples we cannot measure precision or recall — only the rate.

6. What Is Misleading About My Headline Number?

Stratified eval hides the weakest high-volume intent

The eval set has ~24 examples per intent. **ios_update_general** accounts for **18.1% of real traffic** but only 10% of the eval. With a 43.8% error rate, it contributes 7.9% of all real-traffic misclassifications — versus 4.4% in the stratified eval. Reweighting all per-intent accuracies by cluster frequency gives an estimated natural-distribution accuracy of **81.1%** — 2.5 pp below the reported number. Stratified evaluation systematically protects weak high-volume intents from dragging down the headline.

autocorrect_i_bug inflates the score and is largely resolved in production

autocorrect_i_bug (96.0% accuracy) accounts for 17.5% of the eval. This was a specific iOS 11.0–11.1 bug, patched in iOS 11.1.1 released November 9, 2017 — within our dataset's own time window. On current Apple Support traffic this specific bug is largely resolved, meaning a significant portion of our highest-accuracy intent no longer represents active support volume. Removing it and reweighting gives an estimated accuracy of **77.9%** — 4.3 pp lower than reported.

Temporal distribution shift — training and eval share the same bias

The dataset covers October–November 2017 exclusively. The few-shot examples, retrieval pool, and eval set all reflect the same historical moment. The agent was rewarded on iOS 11 vocabulary ("11.0.3", "High Sierra", "question mark box"). On current Apple Support traffic — different devices, different bugs, different vocabulary — the agent faces a cold-start problem. The true production accuracy is unknown and untestable from this dataset alone.

Classification accuracy ≠ customer experience

The 82.2% measures only step 1 of a 3-step pipeline. A correctly classified message that receives a generic "please DM us for more details" reply has not been helped. The most customer-facing metric is **reply relevance (4.31/5)**, the weakest judge dimension. The headline is a necessary but insufficient measure of success.

7. What I Would Do With One More Week

Taxonomy and Classification

- **Sharpen ios_update_general boundary.** Apply rule: use specific intent when a symptom is named; ios_update_general only for general dissatisfaction. Re-label and re-run. Estimated gain: +2–3 pp overall.
- **Add phishing few-shot examples to account_access.** All 6 account_access → general_inquiry failures are vague phishing questions. Examples like "hello is this legit?" would give the LLM the domain prior it is missing.
- **Filter retrieval pool for reply quality.** Exclude threads where all brand replies are under 50 characters or contain only DM redirects. Targets relevance (4.31/5, weakest dimension) by forcing the drafter to use specific Apple guidance.
- **Build a ground truth escalation set.** Hand-label 50 messages as ESCALATE/AUTO_HANDLE. The current 9.5% rate is unvalidated — measuring precision and recall requires labelled ground truth.
- **Expand evaluation metrics.** Measure reply correctness, groundedness (is the reply grounded in retrieved threads?), and escalation precision/recall — not just intent accuracy.
- **Test robustness.** Evaluate on ambiguous, multi-intent, highly emotional, and out-of-distribution messages. Determine when the agent should request clarification or hand off to a human rather than generating a confident but wrong reply.
- **Add production monitoring.** Log latency per component, retrieval similarity scores, escalation rate by intent, intent distribution drift over time, and low-confidence classifications. Enable detection of model degradation before it affects customers.

Dataset: 80,491 AppleSupport threads (Oct–Nov 2017) · Embedding: all-mpnet-base-v2 · Clustering: UMAP + HDBSCAN (106 clusters → 10 intents) · Eval: 242 hand-labelled examples